



Sri Lanka Institute of Information Technology

**IT3021 - Data warehousing and
Business Intelligence**

Assignment 02

IT23816404

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Step 1: Data source for the assignment 2 Data Warehouse Design

The data source for this project is a specialized Retail Operations Data Warehouse based on the SuperStore dataset. This warehouse was designed to transform raw transactional data into a structured format suitable for high-performance analytical processing (OLAP) and business intelligence reporting.

1.1 Data Warehouse Architecture

The data warehouse is implemented using a **Star Schema** architecture. This design simplifies complex queries by separating quantitative business measurements (Facts) from descriptive attributes (Dimensions).

- **Central Fact Table (FactSales):** This table acts as the core of the schema, storing measurable, quantitative data for every retail transaction. Key measures include:
 - **Amount:** The base price of the item sold.
 - **Total_Amount:** The final revenue generated after taxes and discounts.
 - **Ratings:** Customer feedback scores for products.
 - **Sales Count:** A calculated measure to track transaction volume.
- **Dimension Tables:** These tables provide the "who, what, where, and when" context for the facts:
 - **DimCustomer:** Contains demographic information such as Name, Gender, Income, and geographic details (City, State, Country).
 - **DimProduct:** Stores product-specific data, including Category, Brand, and Type.
 - **DimDate:** A time-dimension table used for temporal analysis (Year, Month, FullDate).
 - **DimOrderDetails:** Captures operational logistics like Shipping Method, Payment Method, and Order Status.

1.2 Entity Relationship (ER) Summary

The relationship between the tables is a **one-to-many (1:N)** mapping. Each record in a dimension table (the "one" side) can be associated with multiple records in the FactSales table (the "many" side) via **Foreign Keys** (CustomerKey, ProductKey, DateKey, etc.). This structure ensures data integrity and allows for the rapid "slicing and dicing" of data demonstrated in the OLAP phase.

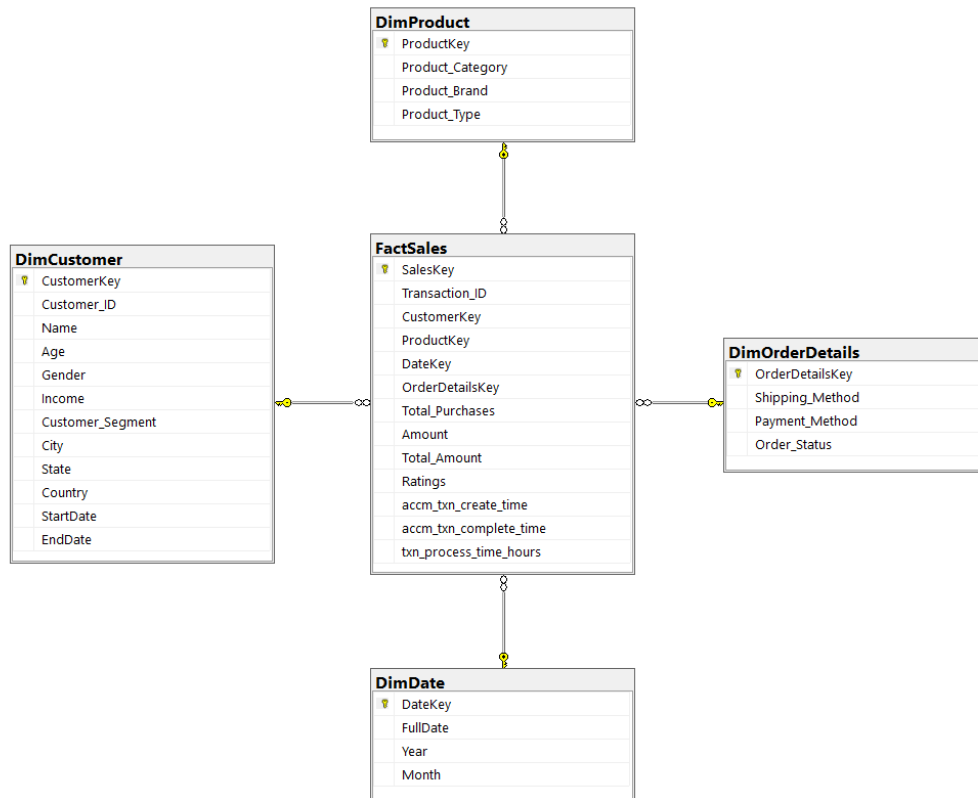


Figure 1: Star Schema for Retail Operations Data Warehouse

Step 2: SSAS Cube implementation

2.1 Project Creation and Configuration

To begin the multidimensional modeling phase, a new SQL Server Analysis Services (SSAS) Multidimensional and Data Mining project was created in Visual Studio, named SuperStore_SSAS. A Data Source (DS_SuperStore_DW.ds) was configured using Windows Authentication to establish a connection with the underlying SQL Server Retail Data Warehouse.

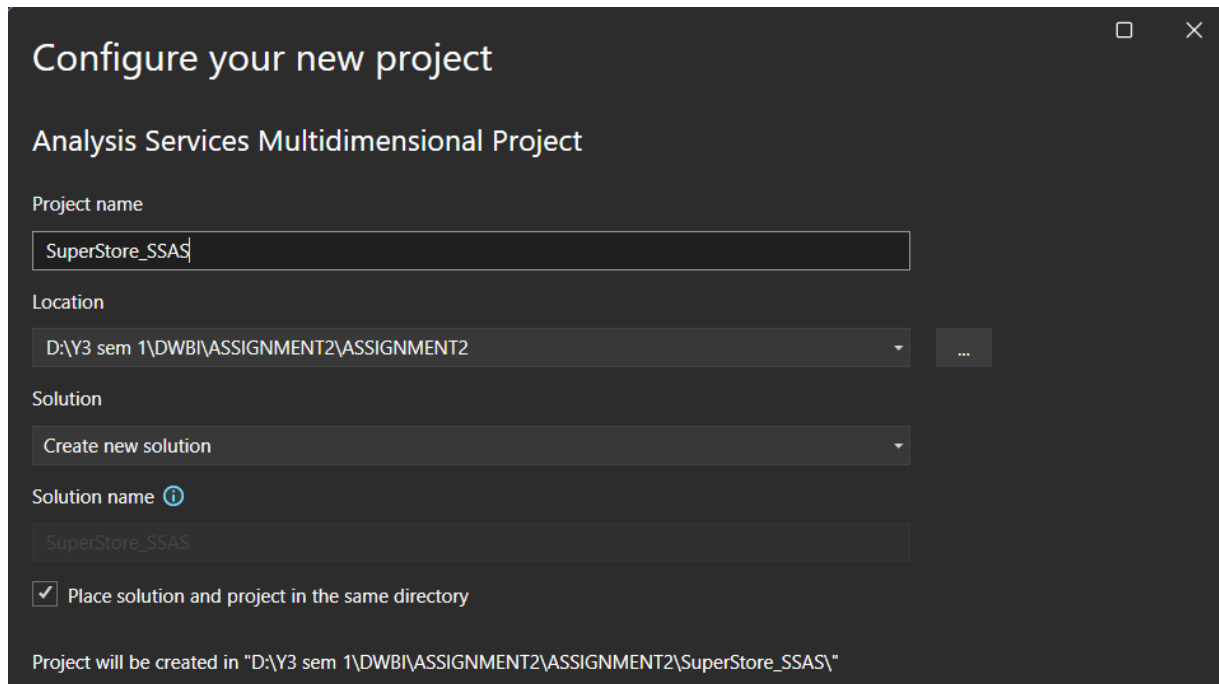


Figure 2.1: Creating a new Analysis Services Multidimensional Project for the SuperStore dataset.

2.2 Established Data Source

Configured a connection to the existing SuperStore data warehouse database (DS_SuperStore_DW.ds) using Windows authentication.

2.3 Built Data Source View

Imported the central FactSales table and all connected dimension tables (Dim Customer, Dim Product, Dim Date, Dim Order Details) from the DW and established the star schema relationships.

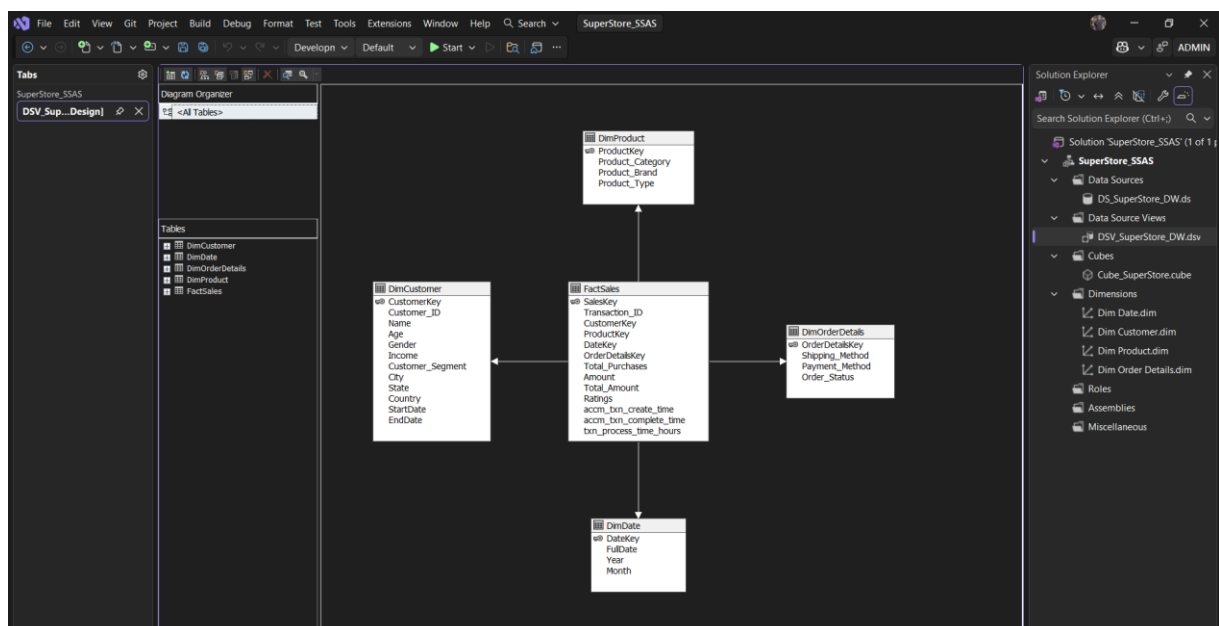


Figure 2.2: Data Source View (DSV) displaying the Star Schema relationships between FactSales and dimension tables.

2.4 Initiated Cube Creation

Used the Cube Wizard with the "Use existing tables" option and selected the FactSales table as the primary measure group.

1. Initiated Cube Creation - Used the Cube Wizard with "Use existing tables" option and selected the fact table as the measure group.
2. Incorporated Dimensions - Added all dimension tables to create the star schema structure.
3. Enhanced Dimensions - Edited each dimension to include all attributes and set appropriate key attributes.

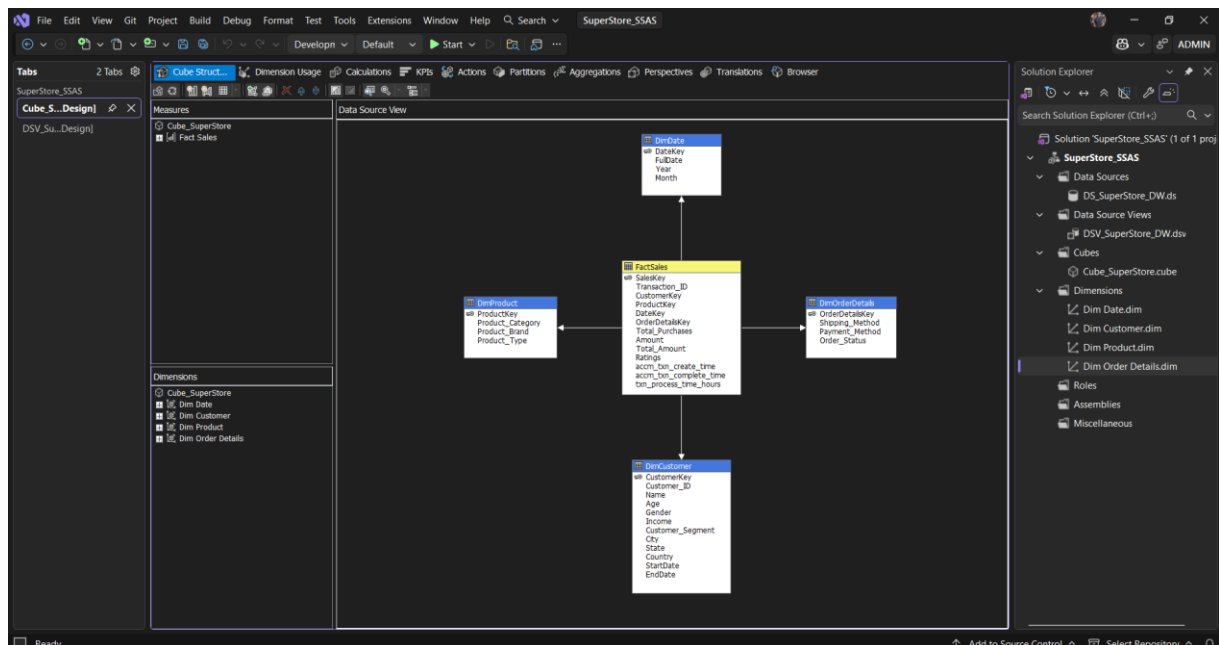


Figure 2.3: Initiating Cube Creation using the SSAS Cube Wizard.

2.5 Incorporated Dimensions

Added all four dimension tables into the cube designer to create the multidimensional star schema structure.

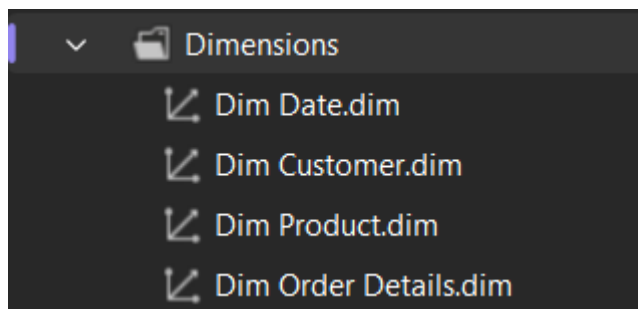


Figure 2.4

2.6 Enhanced Dimensions

Edited each dimension to include all necessary descriptive attributes and set the appropriate key attributes.

2.7 Created Hierarchies

Built user hierarchies within the dimensions to enable drill-down and roll-up analytical capabilities. The following procedural steps were executed in the Dimension Designer to build these structures:

- 2.7.A Opened the Dimension Structure view by double-clicking the target dimension (e.g., Dim Customer.dim) in the Solution Explorer.
- 2.7.B Dragged the relevant attributes from the "Attributes" pane and dropped them into the "Hierarchies" pane in a top-down logical order.
- 2.7.C Renamed the newly formed hierarchy to a descriptive name.
- 2.7.D Configured the Attribute Relationships to ensure correct one-to-many data aggregations and to resolve any duplicate key anomalies.

Using these steps, the following specific hierarchies were implemented:

- **Location_Hierarchy** → Country - State - City
- **Order_Date_Hierarchy** → Year – Month

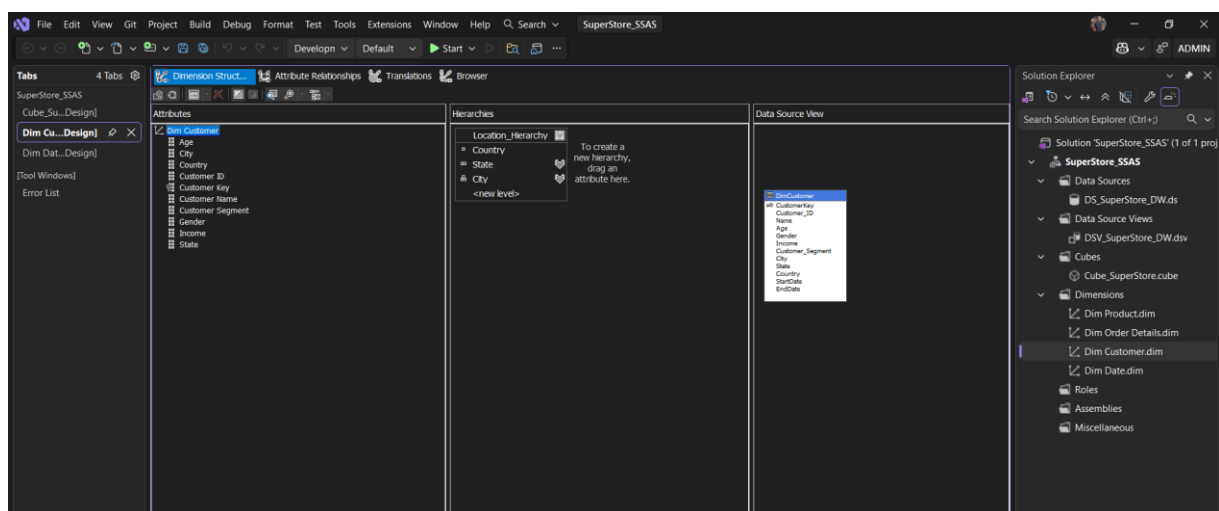


Figure 2.5: Implementation of the Location_Hierarchy within the Customer Dimension.

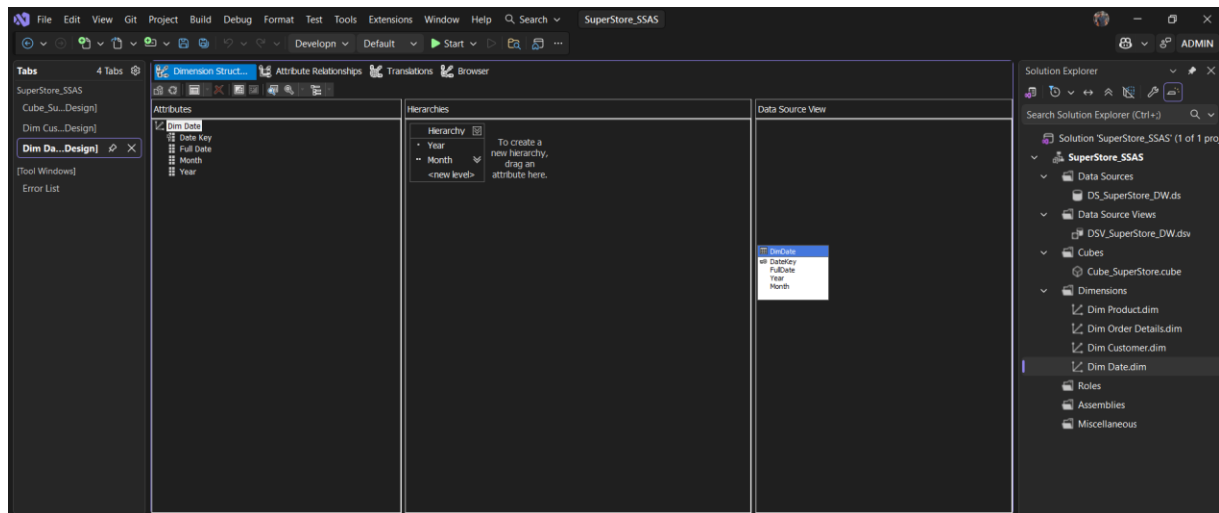


Figure 2.6: Implementation of the Order_Date_Hierarchy within the Date Dimension.

2.8 Verified Relationships

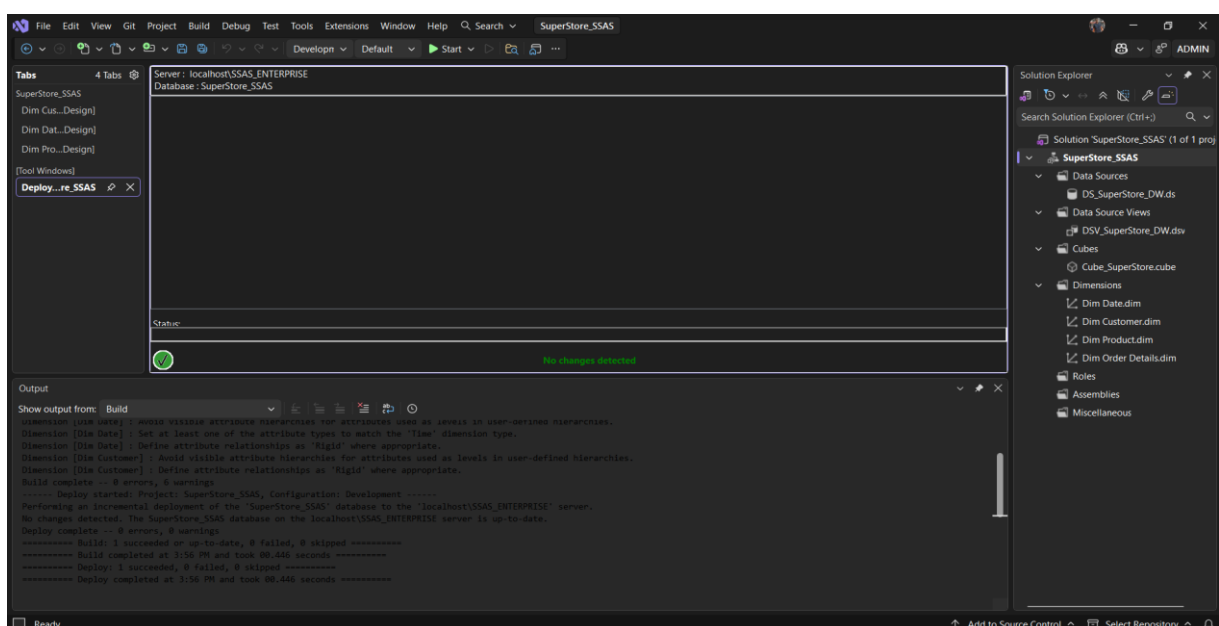
Checked and corrected dimension usage mapping between the FactSales table and the dimension tables to ensure all fact-to-dimension relationships were correctly defined for accurate data slicing.

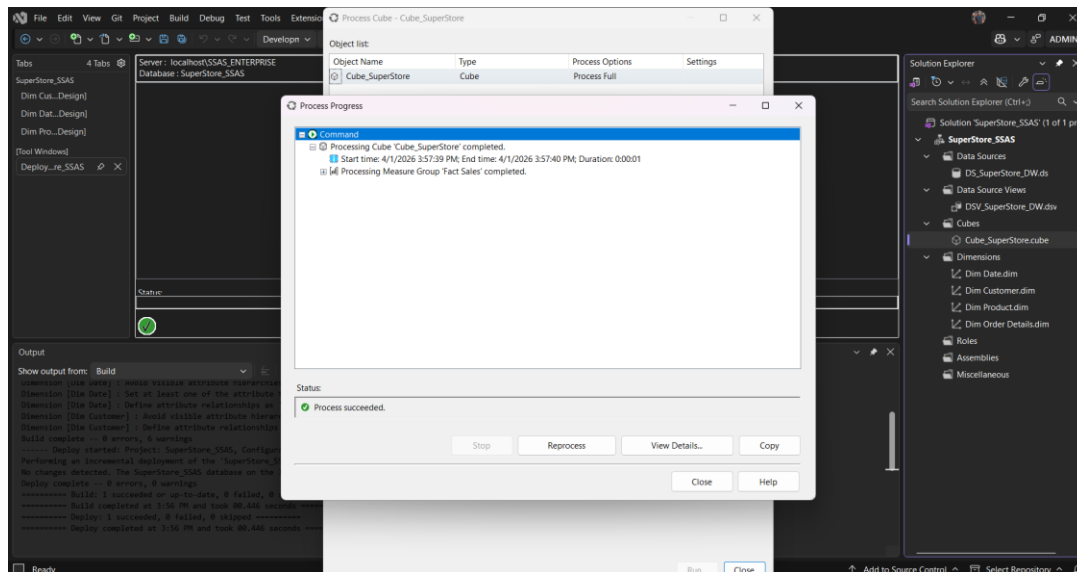
2.9 Deployed the Cube

Configured the project deployment settings to target the local server (localhost). The project was then successfully built and processed using the "Process Full" option to load the data into the multidimensional structure.

2.10 Validated Deployment

Confirmed cube accessibility and successful processing through the deployment progress window, verifying that the cube is fully functional and ready for OLAP operations in client tools like Excel and Power BI.





Step 3: Demonstration of OLAP operations

- 3.1 Connected Excel to SSAS Cube - Used Excel's "Get Data" → "From Analysis Services" feature to establish a live connection to the deployed SuperStore_SSAS cube.
- 3.2 Created PivotTable - Generated a PivotTable report in Excel linked directly to the multidimensional cube for interactive OLAP analysis.
- 3.3 Demonstrated Roll-up - Aggregated data from lower, granular levels to higher summary levels.
 - *Action:* Collapsed the State-level data back into the Country-level total within the Location_Hierarchy.

The screenshot shows an Excel window with a PivotTable in the 'Germany' worksheet. The PivotTable has 'Location_Hierarchy' in the Rows field and 'Total Amount' in the Values field. The data is summarized by country, with Germany having the highest total amount.

Row Labels	Total Amount
Australia	
New South Wales (Australia)	40856378.3
Unknown (Australia)	45153.6117
Canada	41340530
Germany	72296141.6
UK	132122804
USA	115093595
	283953.465
Grand Total	402038556

The PivotTable Fields task pane on the right shows the following configuration:

- Choose fields to add to report:** Fact Sales (Amount, Fact Sales Count, Ratings, Total Amount, Total Purchases). **Total Amount** is selected.
- Drag fields between areas below:**
 - Filters:** (Empty)
 - Columns:** (Empty)
 - Rows:** Location_Hierarchy
 - Values:** Total Amount

This is a close-up of the 'Drag fields between areas below:' section of the PivotTable Fields task pane. It shows the following configuration:

- Filters:** (Empty)
- Columns:** (Empty)
- Rows:** Location_Hierarchy
- Values:** Total Amount

3.4 Performed Drill-down - Expanded hierarchies to analyze granular, detailed data.

3.5

- *Action:* Expanded the Location_Hierarchy at the Country level (Australia) to reveal the underlying State-level sales data (New South Wales, etc.).

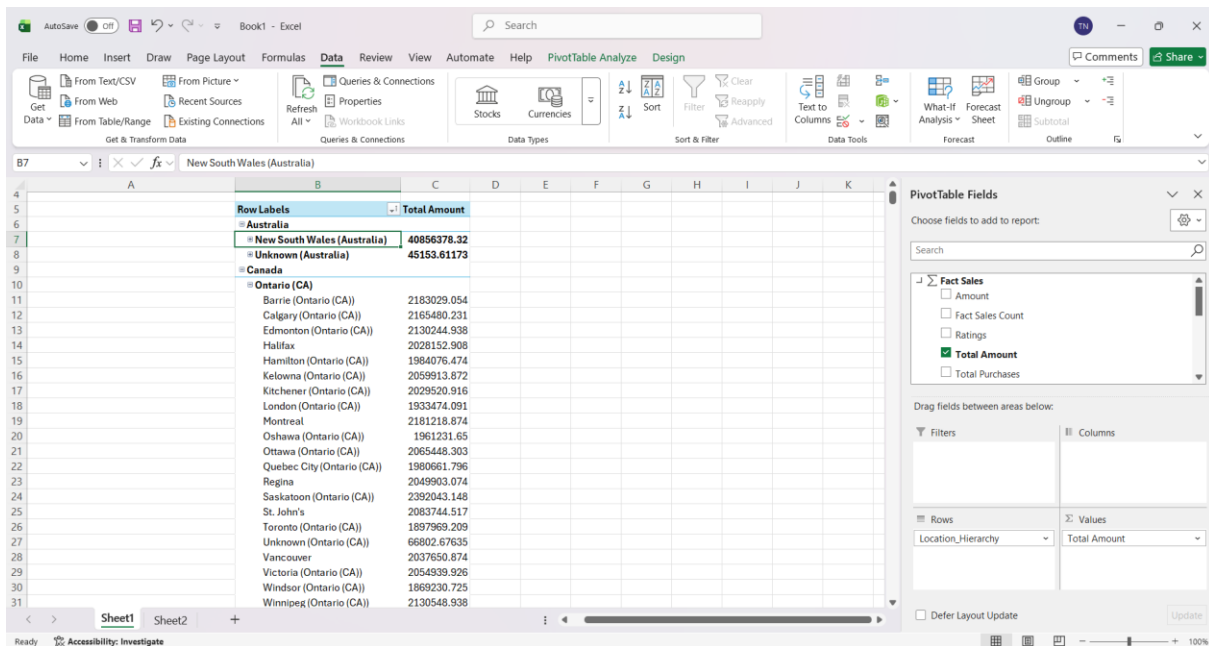
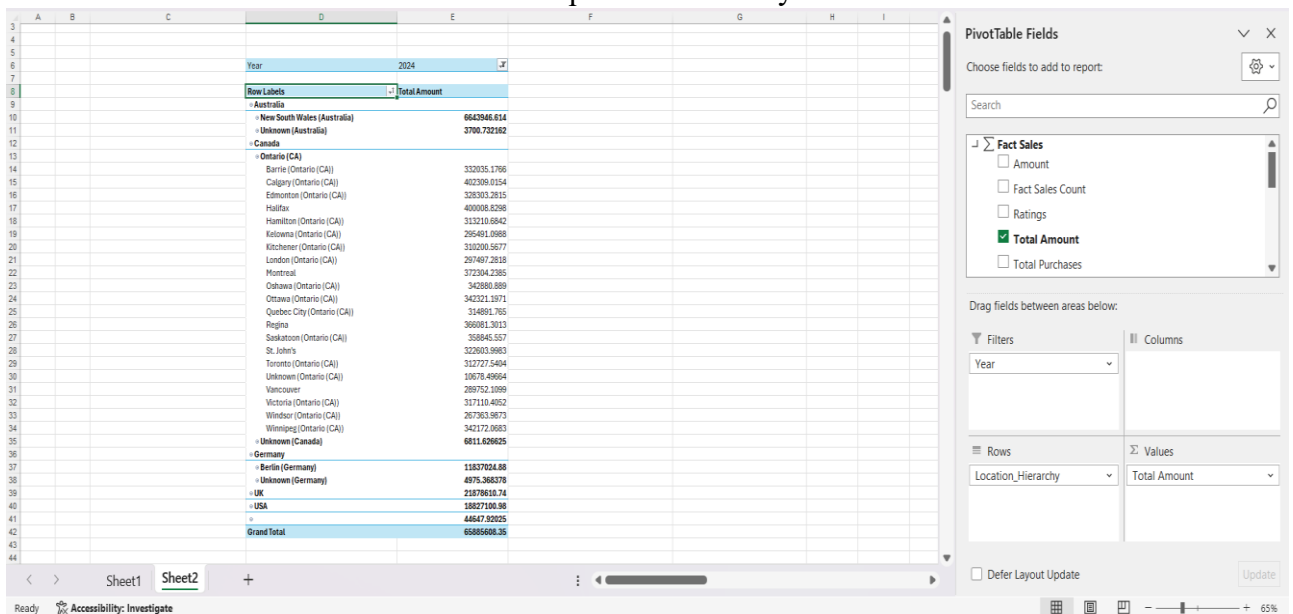


Figure 3.2: Excel PivotTable demonstrating the Drill-Down operation on the Location Hierarchy.

3.5 Applied Slice Operation - Filtered the cube data using a single criterion to view a specific "slice" of the data.

- *Action:* Filtered the entire report to show only records for the **Year: 2024**.



3.6 Executed Dice Operation – Created a sub-cube by applying multiple filters simultaneously.

- *Action:* Filtered the data by **Year: 2024 AND Month: April 2024** to isolate a highly specific chunk of sales data.

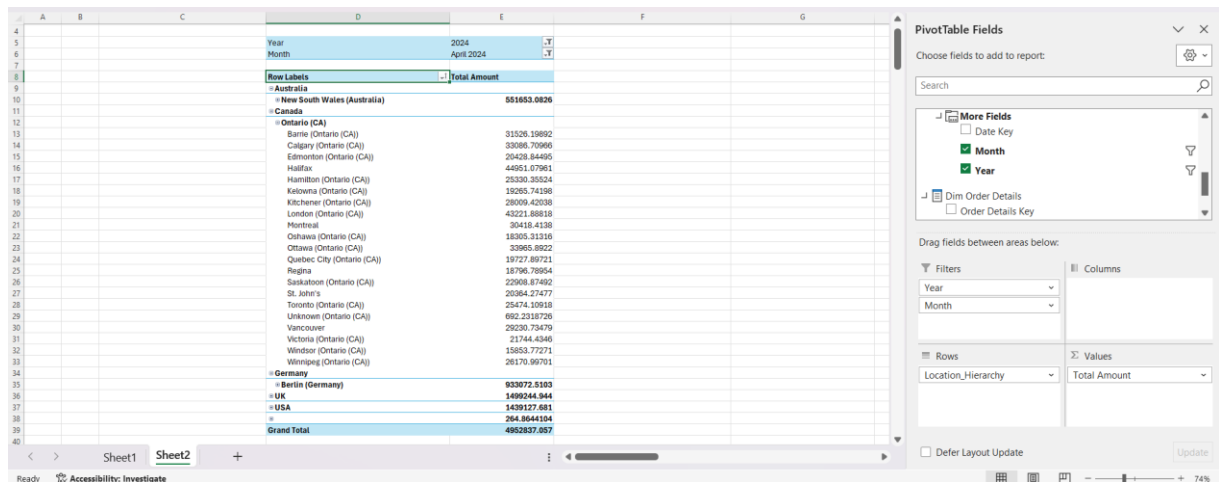


Figure 3.4: Excel PivotTable demonstrating a Dice operation filtering by Year and Month.

3.7 Pivot Functionality - Rotated rows and columns to change the data perspective and provide an alternative analytical view.

- *Action:* Moved the Location_Hierarchy from the Rows axis to the Columns axis, transforming the vertical list of countries into a horizontal layout.

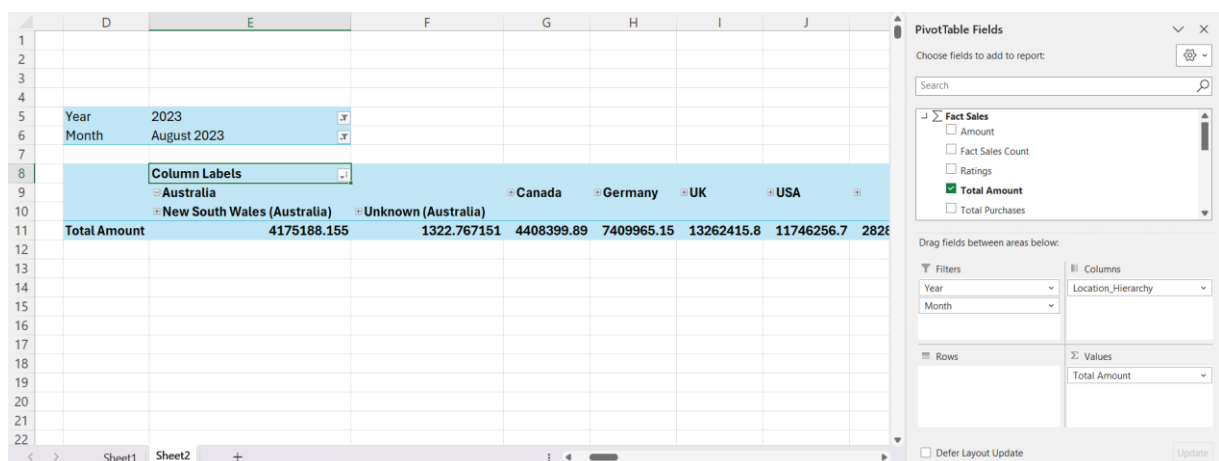
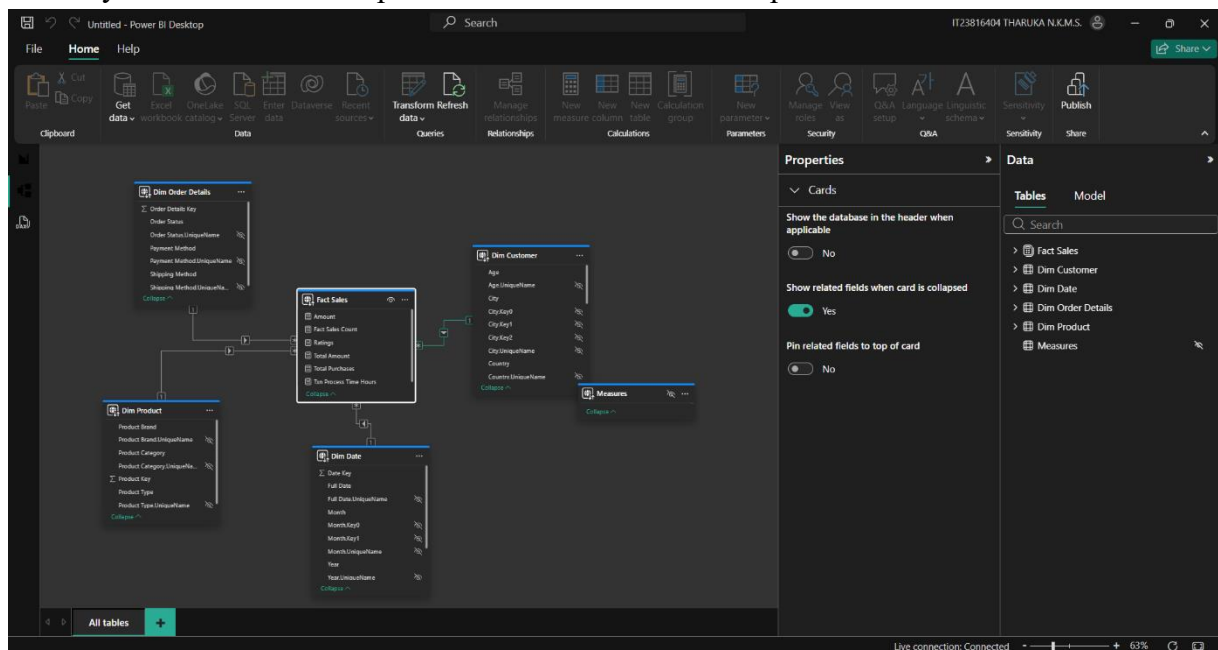


Figure 3.5: Excel PivotTable demonstrating the Pivot operation by rotating the Location axis.

Step 4: PowerBI Reports

The Power BI reports were developed using a Live Connection to the SQL Server Analysis Services database (SuperStore_SSAS), ensuring the visuals automatically reflect the cube's underlying aggregations and structures.

1. Connected to Data Source – Imported fact and dimension tables from the SQL Server Analysis Services Cube into Power BI Desktop.
2. The star schema was already linked with the fact table to the dimension tables using key columns since we uploaded the cube in the last step.



3. Report 1: Matrix Visual

Objective: Display detailed tabular data with row and column groupings.

- **Design:** A Matrix visual was implemented with Product Category, Product Type, and Customer Name nested in the Rows bucket. Year was placed in the Columns bucket.
- **Formatting & Totals:** Subtotals and Grand Totals were enabled to automatically calculate combined sales for each category group and overall performance. Column widths and text alignment were adjusted to ensure a clean, professional layout.
- **Conditional Formatting:** To satisfy the conditional formatting requirement, **Data Bars** were applied to the Total Amount column. Similar to background color shading, these data bars instantly and visually distinguish high-performing sales values (longer bars) from lower sales values (shorter bars) across the grid, allowing for rapid visual analysis of product performance.

Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help

Paste Cut Copy Format painter Clipboard

Get data Excel OneLake SQL Server data Recent sources

Transform Refresh data Queries

New visual Text box More visuals Insert

New visual calculation New measure Quick measure

Product Category	2023	2024	Total
Home Decor	60054610.50	11964082.91	72018693.40
Tools	6559058.87	1357963.93	7917022.79
Lighting	6666104.32	1324224.91	7990329.23
Kitchen	6822133.62	1425169.62	8247303.25
Furniture	13264586.70	2618556.92	15883143.62
Decorations	13378053.37	2694562.94	16072616.31
Bedding	6649259.81	1230791.96	7880051.77
Bathroom	6715413.81	1312812.63	8028226.44
Grocery	73760754.62	14767097.27	88527851.89
Water	27267318.92	5437198.62	32704517.54
Soft Drink	13143920.24	2686536.55	15830456.80
Snacks	6527623.06	1253806.42	7781429.48
Juice	13282941.24	2736092.10	16019033.34
Coffee	6817068.15	1321231.36	8138299.50
Chocolate	6721883.01	1332232.22	8054115.23
Electronics	79090251.48	15420303.08	94510554.55
Total	334679451.52	65671916.96	400351368.48

Product Category	2023	2024	
Home Decor	60054610.50	11964082.91	72018693.40
Tools	6559058.87	1357963.93	7917022.79
Lighting	6666104.32	1324224.91	7990329.23
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Total	334679451.52	65671916.96	400351368.48

Visualizations

Build visual

Filters

Rows

Product Category

Product Type

Customer Name

Columns

Year

Values

Total Amount

Drill through

Cross-report

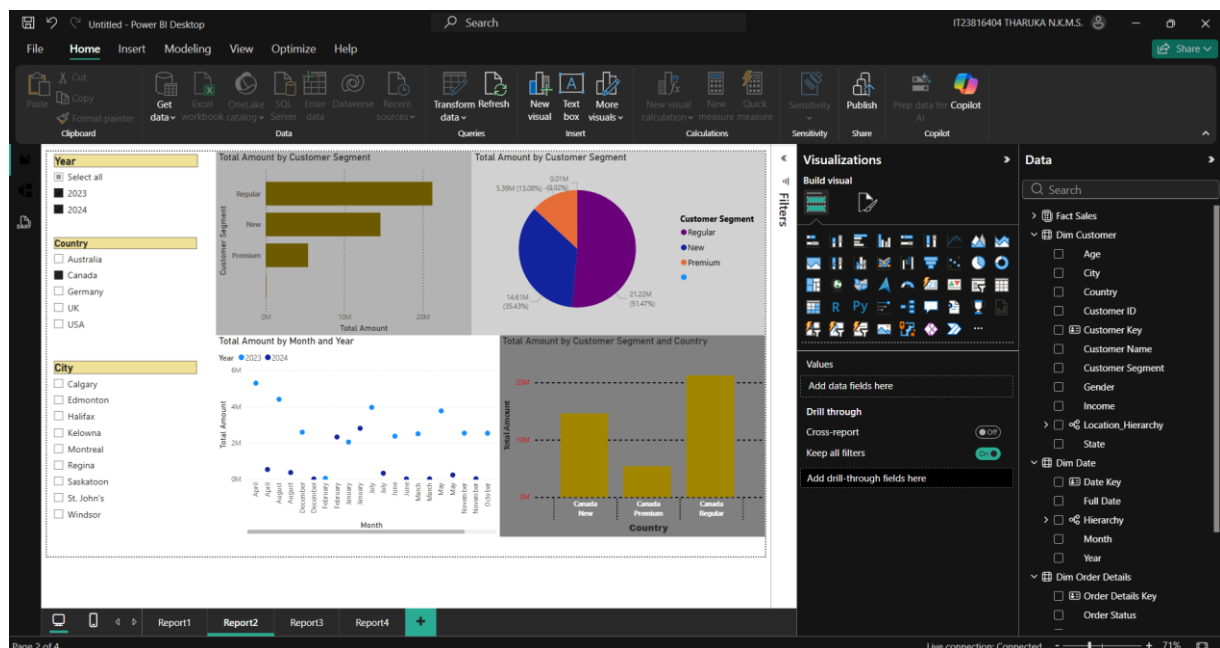
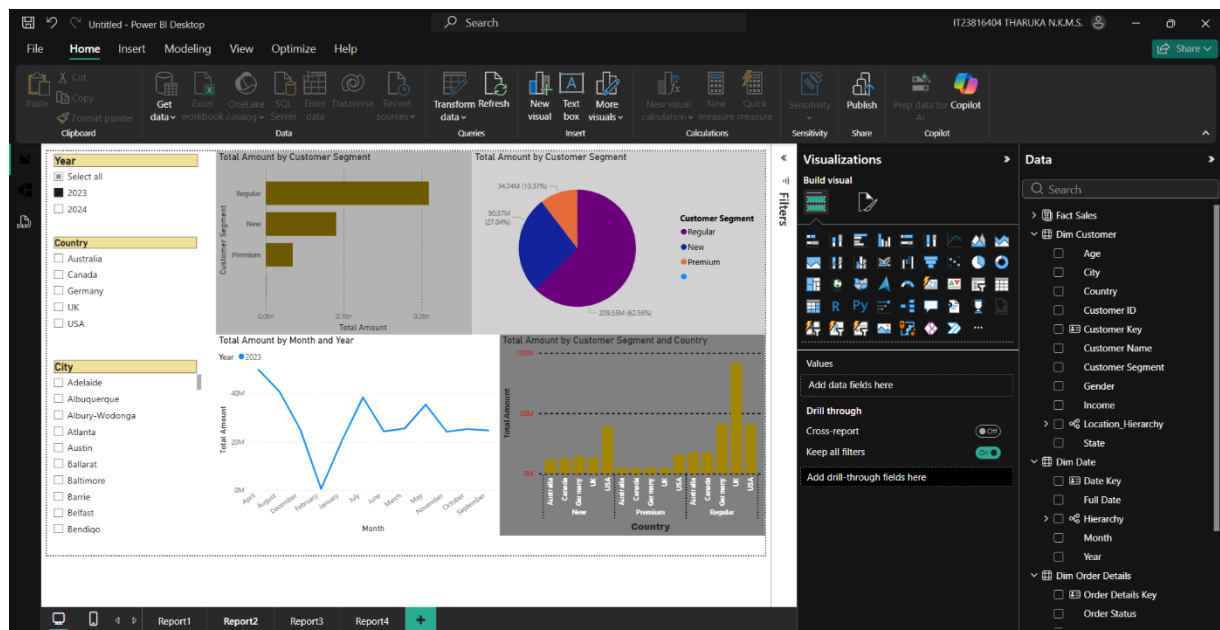
Keep all filters

4. Report 2: Multiple Slicers with Cascading Filters

Objective: To create an interactive dashboard utilizing cascading slicers (parameters) and multiple visual representations to extract dynamic insights from the dataset.

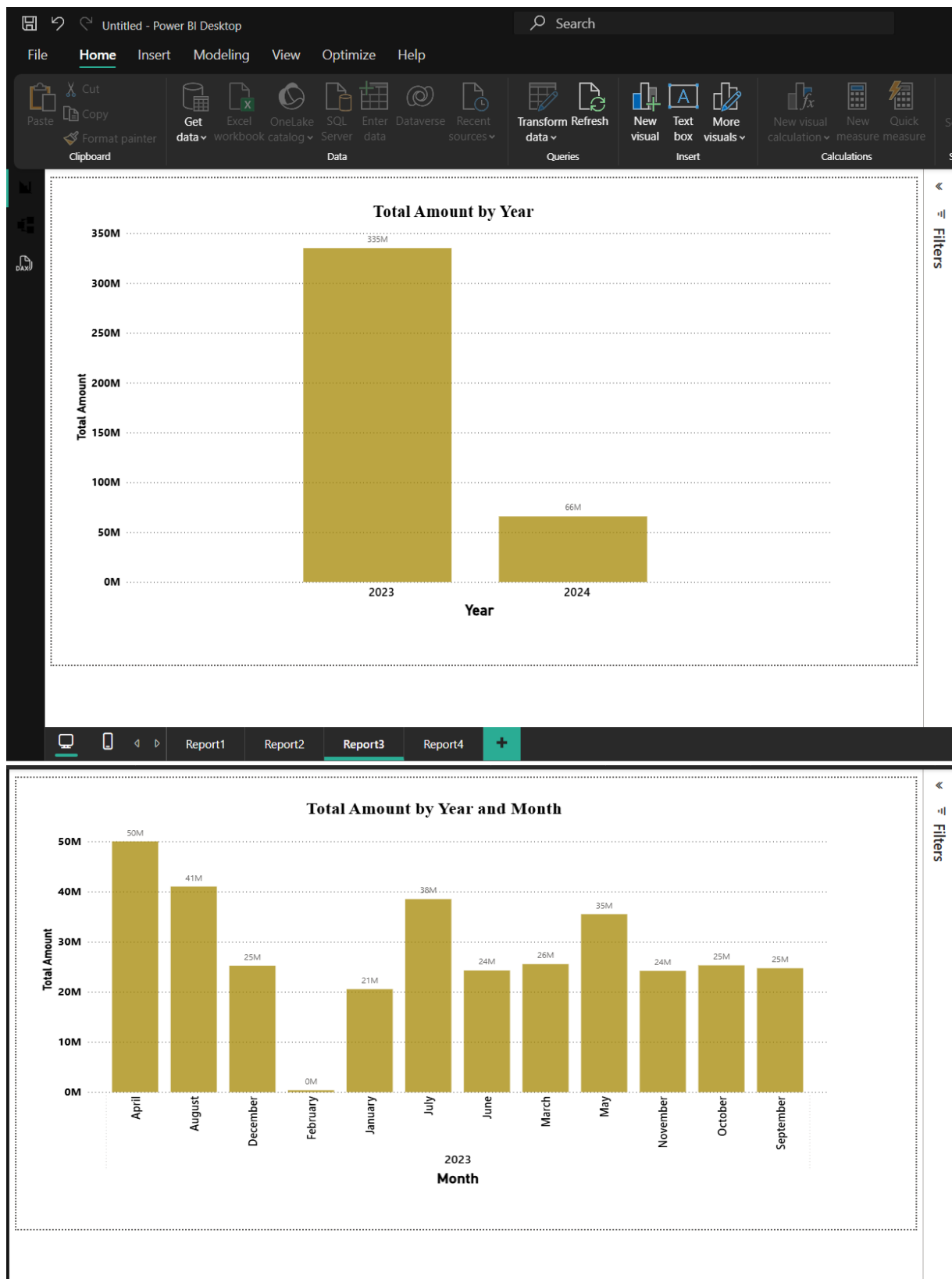
Step-by-Step Implementation:

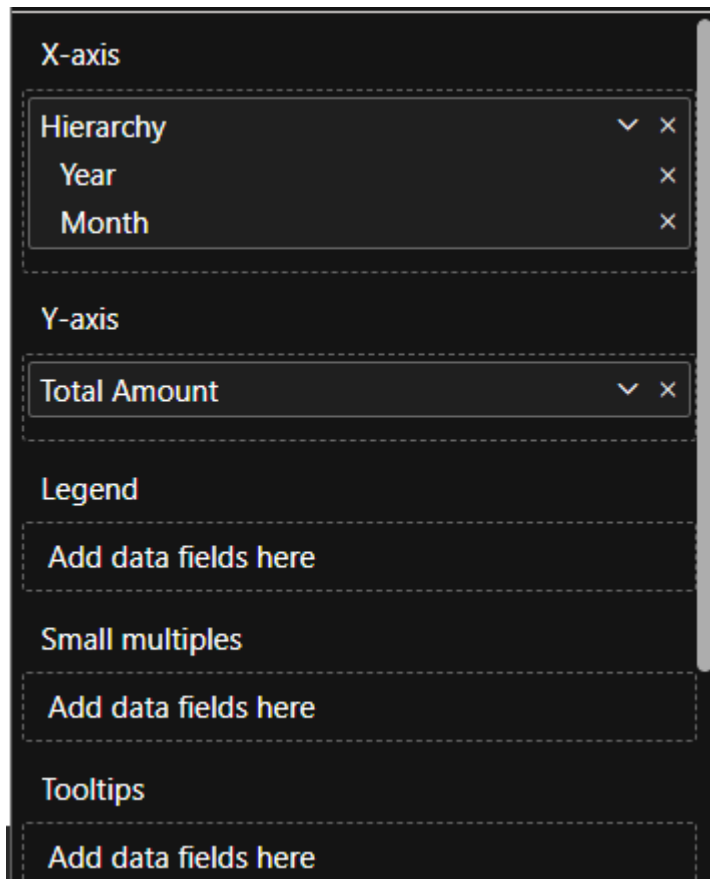
- **Step 1: Implementing Cascading Slicers**
 - Three distinct Slicer visuals were added to the report canvas using fields from the dimension tables: Year (from Dim Date), Country, and City (from Dim Customer).
 - **Cascading Functionality:** The slicers inherently operate with cascading logic due to the underlying cube relationships. When a user selects a specific value in the parent slicer (e.g., selecting "Germany" in the Country slicer), the child slicer (City) dynamically filters to only display cities located within that selected country (e.g., Berlin, Bochum, Bremen).
 -
- **Step 2: Developing Multiple Visuals**
 - To satisfy the requirement for multiple visual insights, four different charts were configured on the canvas to analyze the Total Amount measure:
 1. Clustered Bar Chart: Displays Total Amount by Customer Segment, allowing for a quick comparison of revenue across Regular, New, and Premium segments.
 2. Pie Chart: Illustrates the Total Amount by Customer segment, showing the proportion of sales attributed to the selected geographic regions.
 3. Line Chart: Plots Total Amount by Month and Year to conduct a trend analysis, highlighting sales peaks and troughs across the calendar year.
 4. Clustered Column Chart: Details Total Amount by Customer Segment and Country, providing a deeper cross-tabulation of demographics and geography.
- **Step 3: Dashboard Interactivity**
 - Because the report utilizes a Live Connection to the SSAS Cube, all visuals are highly interactive. Selecting any value in the slicers automatically cross-filters all four charts, allowing the user to drill down into specific regional and temporal insights seamlessly.



5. Report 3: Drill-Down Report.

- Built Hierarchical Visual** – Configured a clustered column chart with drill-down levels to allow hierarchical data exploration. *Date Hierarchy*: Year → Month
- Assigned Values** – Configured the X-axis with the Date Hierarchy and the Y-axis with the Total Amount measure.
- Set up the drill-down functionality** – Enabled the "Drill Down" mode using the single down-arrow visual header icon.
- Demonstrated Exploration** – Tested the functionality by clicking on a specific Year (e.g., 2023) to successfully drill down and reveal the specific monthly sales breakdown for that year.





6. Report 4: Drill-Through Report.

- a) **Built Detail Page** – Configured a specific destination page containing granular details, using a table visual to display Product Brand, Product Type, and Total Amount.
- b) **Set up the drill-through filter** – Added Product Category to the Drill-through fields bucket on the detail page to link the pages together.
- c) **Utilized Main Overview** – Used the primary Matrix visual (Report 1) as the high-level starting point containing the broad categories.
- d) **Demonstrated Drill-Through** – Tested the functionality by right-clicking a specific category (e.g., "Grocery") on the main page and selecting Drill through → Detail Page to view the filtered results.

Power BI Desktop interface showing a report with a table of sales data by Product Category and Year.

Product Category	2023	2024	Total
Home Decor	60054610.50	11964082.91	72018693.40
Copy	6559058.87	1357963.93	7917022.79
Expand	6666104.32	1324224.91	7990329.23
Collapse	6822133.62	1425169.62	8247303.25
Drill down	13264586.70	2618556.92	15883143.62
Expand to next level	13378053.37	2694562.94	16072616.31
Drill up	6649259.81	1230791.96	7880051.77
Include	6715413.81	1312812.63	8028226.44
Exclude	73760754.62	14767097.27	88527851.89
Grocer	27267318.92	5437198.62	32704517.54
Group	13143920.24	2686536.55	15830456.80
Summarize	6527623.06	1253806.42	7781429.48
Unfreeze row headers	13282941.24	2736092.10	16019033.34
Set up a verified answer	6817068.15	1321231.36	8138299.50
Coffee	6721883.01	1332232.22	8054115.23
Chocolate	79090251.48	15420303.08	94510554.55
Electronics	334679451.52	65671916.96	400351368.48
Total			

The right-hand pane shows the 'Visualizations' and 'Data' panes. The 'Visualizations' pane includes options for 'Build visual', 'Filters', 'Rows', 'Columns', 'Values', 'Drill through', 'Cross-report', and 'Keep all filters'. The 'Data' pane shows a search bar and a list of data fields including 'Fact Sales', 'Dim Customer', 'Dim Date', and 'Dim Order Details'.

Power BI Desktop interface showing a report with a table of sales data by Product Brand and Product Type.

Product Brand	Product Type	Total Amount
Bed Bath & Beyond	Bathroom	8,022,172.08
Bed Bath & Beyond	Bedding	7,874,690.94
Home Depot	Decorations	8,132,488.61
IKEA	Decorations	7,921,406.67
Home Depot	Furniture	7,964,407.72
IKEA	Furniture	7,893,201.95
Bed Bath & Beyond	Kitchen	8,238,083.69
IKEA	Lighting	7,987,905.78
Home Depot	Tools	7,910,369.24
Total		71,944,726.67

The right-hand pane shows the 'Visualizations' and 'Data' panes. The 'Visualizations' pane includes options for 'Build visual', 'Filters', 'Values', 'Drill through', 'Cross-report', and 'Keep all filters'. The 'Data' pane shows a search bar and a list of data fields including 'Fact Sales', 'Dim Customer', 'Dim Date', and 'Dim Order Details'.